3.4 Green Transition Dashboard UI/UX

Project plan

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1. Project Working Title + Acronym

Green Transition Dashboard UI/UX (GTD-UX).

2. Background

The Electric Mobility Digital Twin (EMDT) platform collects real-time data from EV fleets, calculating CO_2 savings, energy footprint, and regenerative braking potential based on road topography. 3.4 Green Transition Dashboard UI/UX project goal is to make a well-designed user interface through which operators can monitor these KPIs and explore optimized routes.

Scope:

- In scope: UX research, wireframes, high-fidelity Figma prototype, map view UI, usability testing, design system, final report
- Out of scope: backend/API development, live data integration, mobile app

3.Project Objectives and Scope

Main objective: Design a usable, clear dashboard UI/UX prototype for the EMDT Green Transition Dashboard, enabling users to monitor CO_2 savings and energy footprint KPIs and explore topography-based EV route optimization visually.

Sub-objectives:

- Literature reviews and benchmarking of sustainability and EV dashboards
- Define user personas and key use cases
- Make low-fidelity wireframes for all core components in Figma
- Develop high-fidelity interactive prototype in Figma
- Build interactive Dashboard according to prototype in React
- Do usability testing (peer testing + UI/UX testing methods)
- Improvement according to usability testing
- Deliver design system and final project report

Research question: How can real-time dashboard UI/UX effectively communicate CO_2 savings, energy footprint KPIs and support users' decision-making in the EMDT platform.

4. Project organisation and management

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	Deliverable
Literature Review & Research										Research notes
UX Research & Personas										Personas, use cases
Wireframing (Low-fi)										Wireframe set
High-Fidelity UI Design (Figma)										Figma prototype
Build the Dashboard according to prototype										Initial version demo
Usability Testing & Iteration										Test report
Improvement according to usability testing.										Final version
Report, Design System & Presentation										Report + slides

5. Stakeholder and Risk Analysis

Student **Unubold Luvsandagva** – Responsible for all deliverables
Supervisor **Mira Vorne** – academic guidance and feedback
EMDT Project Team – Client

Risk analysis:

Risk	Likelihood	Impact	Mitigation
Tool unfamiliarity (Figma, Leaflet.js)	Medium	Medium	Week 1-2 includes tool exploration
Limited real users for testing	High	Medium	Peer testing + heuristic evaluation
Time management	Medium	Medium	Weekly task time management
Changing requirements from EMDT team	Low	Medium	

6. Challenge and Opportunities

Challenges:

- Translating complex EV/energy data into intuitive visuals without oversimplification
- Limited access to real end-users; mitigated by peer testing and heuristic evaluation
- Solo workload - all roles fall on one person; managed through structured scheduling and weekly review

Opportunities:

- Real client context (EMDT) gives the design work genuine practical value
- Builds a strong portfolio case study in a specialist, high-demand UI/UX niche

6.Summary

GTD-UX is a UI/UX design project within LAB REDI 2026, aiming to design a clear and usable dashboard prototype that communicates real-time CO_2 savings, energy footprint KPIs, and topography-based EV route optimization for the Electric Mobility Digital Twin platform - delivering wireframes, a high-fidelity Figma prototype, and a design system.